DARKWOODS FOREST CARBON PROJECT

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Project Title	Darkwood Forest Carbon Project
Version	v1.0
Date of Issue	20-September-2019
Project Location	Canada, British Columbia
Project Proponent(s)	Nature Conservancy of Canada Rob Wilson rob.wilson@natureconservancy.ca +1 800 465 0029
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Project Lifetime	01 April 2008 – 01 April 2107; 100-year lifetime.

Monitoring Period of this Report	01 January 2017 – 31 December 2018
History of SD VISta Status	New SD VISta project; initial verification.
Other Certification Programs	Verified Carbon Standard Project ID: 607. Climate, Community & Biodiversity Standard Project ID: 607..
Expected Future Assessment Schedule	Initial Validation/Verification: Fall 2019.

CONTENTS

1 SUMMARY OF SDG CONTRIBUTIONS

Table 1: Summary of SDG Contributions

Row number	Quantitative Project Contributions during Monitoring Period	Contributions during Project Lifetime	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Section Reference	Claim, Asset or Label
1)	Carbon Emissions Reductions (2017-2018): 871,508 tCO_2e.	Carbon Emissions Reductions (project lifetime): 4,383,123 tCO_2e.	13.0	Tonnes of greenhouse gas emissions avoided or removed	Increase	VCS & CCB validation report.	SD VISta - labeled VCU
2)	Forest Biomass (2017-2018): a net ~1,859 ha of avoided harvesting in this monitoring period.	Forest Biomass (project lifetime): net ~7,941 ha of avoided harvest to date.	15.2	15.2.1	Increase	VCS validation report	Claim
3)	Water: water is qualitatively assessed as net positive in this monitoring period.	Water: water is qualitatively assessed as net positive over the project lifetime.	6.1		Increase		Claim

4)	Biodiversity: see Table 10.	Biodiversity: see Table 10.	15.5		Implemented activities to increase.	CCB Validation Report	Claim

2 PROJECT DESIGN

2.1 Project Objectives, Context and Long-term Viability

2.1.1 Summary of Project Sustainable Development Objective(s)

The Nature Conservancy of Canada (NCC) acquired the fee simple 54,792 ha (135,394 acre) Darkwoods property near Creston, British Columbia, Canada from the Pluto Darkwoods Corporation in April of 2008, with the objective of managing the land for biodiversity, climate, and community benefits. The Darkwoods Forest Carbon Project was developed and Validated and Verified to the Verified Carbon Standard (VCS), AFOLU Methodology VM0012 as a 100 year Improved Forest Management (IFM) – Logged to Protected Forest (LtPF) carbon project on April 21, 2011. The project was Validated and Verified to the Climate, Community, and Biodiversity Standard (CCB) on December 15, 2016 and February 27, 2018, respectively.

Prior to the project, the Darkwoods property was managed for timber production as a private timberland, and was put up for sale under a valuation based on timber and other development values. The Darkwoods Forest Carbon Project involves the acquisition and perpetual management of the property for long term conservation and improvement of ecological function and values through the protection of the forest and various conservation-based forest and land management activities.

The primary objectives of the Darkwoods project is to actively manage and protect the property to achieve climate benefits, biodiversity benefits, water benefits, and other ecosystem services benefits; while retaining substantial net benefits for local communities.

The Darkwoods project achieves net GHG emission reductions through the avoidance of emissions due to conventional logging in the baseline scenario, along with additional carbon sequestration by retaining additional forest biomass and older forests across the project area over time.

The project achieves an exceptional level of biodiversity protection and enhancement through the protection and management of a regionally significant land area with fully functional natural ecosystems and critical habitat for key endangered species.

The project achieves a significant level of water benefits that improve water quality, quantity, timing of flows, and important aquatic habitat improvements over the baselines conditions through the retention and management of riparian habitat, watershed-level forests and ecosystems, and road and access management.

The project area has no communities within or dependent on the project area, however the project achieves net socioeconomic benefits for nearby local communities through limited forest and conservation management and research activities, ecosystem services provisioning, recreational opportunities, and improving community amenity values. The management of the property for these purposes does not generate material levels of net revenue, and the sale of carbon offsets and other ecosystem services benefits are critical to funding significant ongoing property conservation management and ownership costs.

The major objectives of the project are to protect and enhance the long-term function of the natural ecosystems on the Darkwoods property, including:

1. Protecting and enhancing biodiversity through the conservation and management of fully functioning natural ecosystems and key species habitat over time, including:
    a. Restore and maintain Mountain Caribou habitat and movement areas.
    b. Restore and maintain hydro-riparian ecosystem integrity, riparian forest structure and function, fish, and hydrologic functioning.
    c. Restore and maintain stand structure within the range of natural variability for Dry ICH forest.
    d. Restrict human access to core Mountain Caribou and Grizzly Bear habitat.
    e. Maintain or restore Grizzly Bear habitat and movement areas.
    f. Restore and maintain old-forest attributes in old-growth and recruiting cedar-hemlock forests.
    g. Maintain the integrity of rare ecosystems and habitat features.
    h. Restore and maintain full suite of naturally occurring species and ecosystems.
    i. Control noxious weeds and other problematic invasive species.
2. Protecting current carbon stocks and future carbon sequestration potential of the ecosystems across the property, including:
    a. Reducing timber harvesting to low levels related to long term ecosystem and habitat management, natural disturbance risk reduction, salvage, or other conservation based objectives
    b. Monitoring property carbon stocks over time
3. Protecting fully functional ecosystems that provide important services and amenities to local communities, while providing diverse economic opportunities; including:
    a. Providing ongoing direct economic opportunities through project activities, expenditures, and operations;
    b. Protecting the property to retain or enhance ecosystem services provisioning, retaining visual quality and other features contributing to regional wildland amenity values, and allowing limited low impact backcountry recreational and other use opportunities as appropriate
    c. Supporting research, monitoring, and other diverse economic opportunities across the property, as feasible

d.

2.1.2 Description of the Project Activity

Project Activity 1 – Conservation of current forests and ecosystems:

The baseline scenario involves timber harvesting to maximize economic opportunities and achieve standard industry returns on investment; which results in a baseline scenario of harvesting ~300,000 m^3/year for the first 15-20 years, followed by periodic harvesting as new stands mature to harvest age over the project lifespan. The primary objective and activity of the project is to avoid this timber harvesting regime and manage the property for conservation purposes. This reduction in timber harvesting results in:

1. Climate impacts: avoidance of GHG emissions resulting from the high level of timber harvesting in the baseline scenario over the project timeline
2. Community impacts: protection of key ecosystem services provisioning related to water quality, air quality, biodiversity, etc. Retention of regional wild land related amenities.
3. Biodiversity impacts: avoidance of habitat modification and operational impacts of logging in the baseline scenario. Retention of large-scale areas of fully functional natural ecosystems and habitat.

Project Activity 2 – Conservation-based forest management:

NCC intends to undertake various conservation management activities, including low levels of timber harvesting, road maintenance and deactivation, silviculture, and various other property management activities to achieve and enhance long term conservation management objectives. The largest impact on forest conditions and GHG emissions during the project will result from an expected low level of timber harvest of approximately 10,000 m^3/year.

1. Climate impacts: avoidance of GHG emissions related to larger scale timber harvesting in the baseline scenario over the project timeline
2. Community impacts: protection of key ecosystem services provisioning related to water quality, air quality, biodiversity, etc. Retention of regional wild land related amenities. Retain and provide a set of limited direct economic opportunities for forest products and forest management in the local communities
3. Biodiversity impacts: avoidance of habitat modification and operational impacts of industrial logging in the baseline scenario, including road deactivation. Retention of large-scale areas of fully functional natural ecosystems and habitat.

Project Activity 3 – Project monitoring.

The project undertakes a variety of ongoing forest and ecosystem inventory activities to monitor property management objectives. In addition to maintaining a comprehensive forest GIS inventory system, NCC has installed an network of permanent plots to measure the current and ongoing carbon balance of the Darkwoods forests over time.

1. Climate impacts: the inventory activities have no material impact on climate other than as related to successfully implementing the carbon project itself.

2. Community impacts: monitoring activities provide direct economic opportunities to local and regional people.
3. Biodiversity impacts: the activity has no material impact on biodiversity other than as related to successfully implementing the project itself.

Project Activity 4 – Biodiversity project monitoring.

NCC has designed and implemented a robust conservation plan and monitoring program for Darkwoods. The Darkwoods property participates in a series of scientific research projects related to key habitat, mapping, threatened species, and other elements.

1. Climate impacts: the activity has no material impact on climate other than as related to successfully implementing conservation plans on Darkwoods.
2. Community impacts: monitoring activities provide direct economic opportunities to local and regional people. As part of the results of conservation planning, NCC has implemented a public access permitting system, along with a series of user lease agreements and other access planning processes that provide opportunities for community members to interact with the Darkwoods property.
3. Biodiversity impacts: the activity has no material impact on biodiversity other than as related to successfully implementing the conservation management of Darkwoods itself.

2.1.3 Implementation Schedule

Date	Milestone(s) in the Project's Development and Implementation
Apr. 1, 2008	Project start date - Darkwoods property acquisition closing date and start of project activities.
Apr. 21, 2011	VCS project validation
Apr. 21, 2011	VCS project verification for periods 2008, 2009, 2010
Mar. 25, 2014	VCS project verification for periods 2011, 2012
Dec. 15, 2016	CCB project validation
Feb. 27, 2018	VCS project verification for periods 2013, 2014, 2015, 2016
Feb. 27, 2018	CCB project verification for periods 2013, 2014, 2015, 2016
Dec., 2019	Expected VCS & CCB verification for periods 2017, 2018
Jan., 2020	Expected VCS & CCB verification for period 2019
2019	SD VISta project validation undertaken
2019	SD VISta initial project verification for period 2017, 2018
Annually/ongoing	Project implementation, property management activities
Annually/ongoing	Project area monitoring

2.1.4 Project Proponent

Organization Name	Nature Conservancy of Canada (NCC)
Role in the Project	Project Proponent
Contact Person	Rob Wilson
Title	Director - Carbon Finance & New Conservation Strategies
Address	Suite 410 – 245 Eglinton Avenue East, Toronto, Ontario, Canada M4P3J1
Telephone	+1 800 465 0029
Email	rob.wilson@natureconservancy.ca

2.1.5 Other Entities Involved in the Project

Organization Name	RainCloud Forests
Role in the Project	Project Developer
Contact Person	Mike Vitt
Title	CEO
Address	3516 Skylark Loop, Bellingham, Washington, USA 98226
Telephone	+1 206 307 7096
Email	mike.vitt@raincloudforests.com

2.1.6 Project Location

The Darkwoods property is a 54,792 ha (135,394 acre) contiguous parcel of fee simple private property in south eastern British Columbia just north east of the municipality of Creston. The Darkwoods property is bounded by Kootenay Lake on the east and various crown and private land on the other property boundaries. A significant in-holding property in the center of the Darkwoods property has been recently acquired by NCC and the Province of British Columbia (closed 2019) with management plans under development that should be similar/compatible with the project management activities on Darkwoods.. that is owned and managed by the Wynndel Box and Lumber Company, with access rights through the Darkwoods property. The property and project boundaries are shown in Figure 1, and Figure 2.

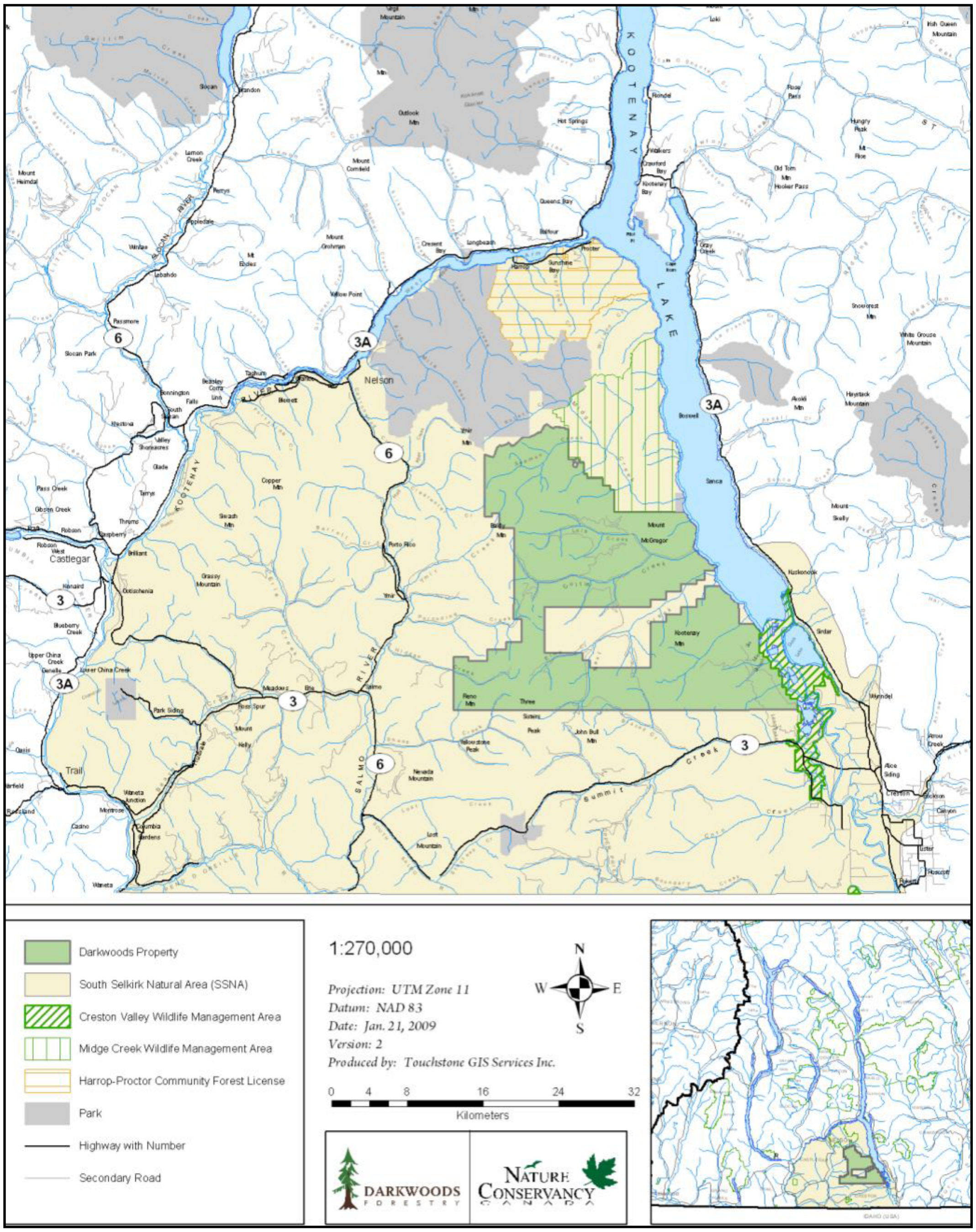

Figure 1. Darkwoods Property Boundary and Project Area Overview Map

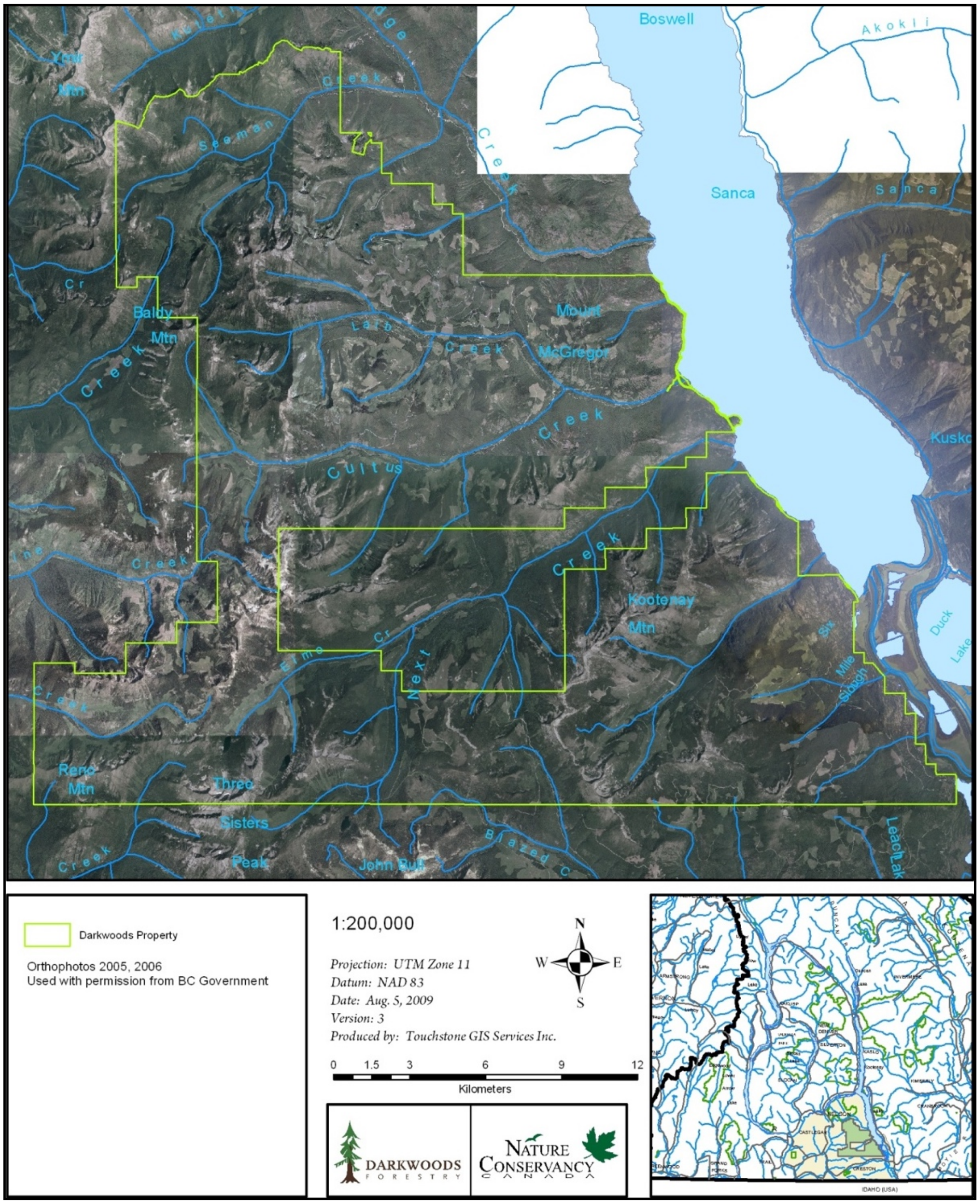

Figure 2. Darkwoods Property Ortho Photo Mosaic

2.1.7 Project Description Deviations

As the initial verification is being conducted simultaneously with the PD validation, there are no deviations for this monitoring period.

2.1.8 Threats to the Project

The risks and threats to the sustainable development are outlined in detail in the VCS Non-Permanence Risk Assessment documentation for the 2017-2018 monitoring period. There are no additional threats to the project specifically related to the SD VISta elements in the project.

Summarized, the project area generally has the spatial scale and biological, terrain, and climatic diversity to be low risk for significant loss to carbon, biodiversity, and community values in the project. Forest fires do occur periodically on the property and are likely one of the higher direct risks to climate and biodiversity impacts. However, fires are rare in this wet/moist bioclimatic zone, while the steep and diverse terrain of Darkwoods makes large scale fire damage rare. NCC has a staffed field office and constant presence on the property, and has a fire fighting agreement with the British Columbia Ministry of Forests who have significant firefighting assets in nearby Creston, B.C. Collectively these features strongly reduces the risk of fires and significant damage to forest biomass on the property.

Human induced threats are also rare on the Darkwoods property due to its relatively isolated location, gated road access controls, and NCC's recreational use permitting system; along with being located in a fully developed region with excellent judicial and enforcement systems. The property is owned by the largest conservation group in Canada (a not-for-profit organization) and is protected by perpetual conservation easements and the Canadian Federal Eco-gifts Program participation agreement, which collectively provide excellent ownership risk mitigation.

During this monitoring period, two naturally caused forest fires occurred on the property in 2018. These fires resulted in a variety of burn severities covering ~2,447 ha of the project area. The resulting estimated carbon stock changes have been accounted for in the ex-post calculation of emissions reductions for the project during this monitoring period. This did not result in a VCS loss event, as emissions related to these fires were simply netted to current year emission reductions, however, Verra has been notificed of the occurances by email. The fires are consistent with the level of frequency and severity noted in the VCS Non-Permanence Risk Assessment Tool. These fires periodically occur in these ecosystems, and these occurrences did not have a material adverse impact on the net positive benefits of the project for climate, biodiversity or community.

2.1.9 Benefit Permanence

The Darkwoods property is owned and managed for conservation purposes in perpetuity by the Nature Conservancy of Canada. The property also participates in the Canadian Eco-Gifts Program that provides legally binding perpetual permanence protection and enforcement mechanisms to insure the property is managed for conservation purposes in perpetuity.

The result is the benefits to climate, community, and biodiversity will be retained in perpetuity as the property retains, regrows, and enhances fully functional ecosystems across the property. There has been no changes impacted benefit permanence during this monitoring period.

2.2 Stakeholder Engagement

2.2.1 2.2.1 Stakeholder Consultation and Adaptive Management

NCC has undertaken a significant and ongoing effort to communicate with the local community generally on a periodic basis, and specifically with any directly interested stakeholders on an as-needed, or at-request basis. Additionally NCC has an active permanent presence in the local community and region, along with active webpages and social media accounts, such that any other interested stakeholder has the opportunity to communicate with NCC through direct contact and in periodic public meetings.

There are a few examples of local community members who are active on the project area, but who are not negatively impacted by the project activities, including:

- There are currently 27 non-permanent recreational cabins/trailers with associated lake moorages located on a small area on the eastern edge of the property along Kootenay Lake, known as Tye. NCC has permitted a continuation of the pre-existing seasonal recreational cabin usage through a process of annual rental agreements. NCC retains a year-round caretaker to oversee and care for this Tye recreation area. NCC communicates directly with the Tye community on an as-needed/ongoing basis.
- There are a few (1-2 active) small (3-4 workers) exploration and mining/prospecting operators occasionally active on the property (but no productive mines). These operations have a minimal surface impact (<1 hectare) and their impacts on carbon emissions are immaterial. NCC communicates directly with these operators on an as-needed basis.

During this monitoring period NCC has periodically communicated with these interested stakeholders through phone, email, and letter correspondence. No issues of note have arisen during this period.

In general, the local interested stakeholders that actively engage with NCC related to the Darkwoods property and that may be directly or indirectly impacted by the project include:

- Project-based contracting opportunities for local road construction, maintenance, timber harvesting, and/or silvicultural activities within the project
- Project based forest management consulting for inventory, planning, monitoring, or other project related work
- Backcountry recreational users
- Researchers undertaking scientific studies

NCC communicates periodically with these interested stakeholders as necessary or requested on an annual or periodic basis.

In addition, there are regional groups, associations and members of the public in Nelson, Creston, and the surrounding areas that occasionally express interest in conservation initiatives, research, and other general happenings on the Darkwoods property and the surrounding area. NCC meets and communicates with these groups on a periodic basis or upon request related to the carbon project, project activities, and plans on Darkwoods; along with other regional NCC activities.

The property sits within the traditional territory of the Ktunaxa First Nation. Generally private property rights in Canada are respected by First Nations, who primarily focus on land use rights and territorial issues on the larger Crown land controlled by the province. The Ktunaxa are important regional stakeholders who have an interest in cultural and environmental considerations on all areas of their territory, including Darkwoods; however, there is no material level of ongoing use, livelihood activities, or special cultural values on the project area and within the project zone that are materially impacted by the project activities (i.e. Choquette, 2010). NCC has a strong and active working relationship with the Ktunaxa and is engaged on several regional initiatives of mutual interest.

NCC has communicated with interested stakeholders on a period and ongoing basis throughout this monitoring period, as shown in Table 2 and Table 3 .

Table 2. Examples of interested stakeholders (recreational groups) engaged with NCC during the monitoring period.

Recreational Groups
Kootenay Mountaineering Club
Beaver Mountain Snowmobile Association
Kokanee Snowmobile Club
Nelson Snowgoers Club
Creston Rod and Gun Club
Association of BC Snowmobile Clubs

Table 3. Scientific researchers engaged on Darkwoods during this monitoring period.

Scientific Researchers	Topic	Years
Michael Proctor	Grizzly Bears	2013-2016
Adrian Leslie	Whitebark pine	2013-2014
F&W Compensation Program	Bull Trout Redd Counts	2017-2019
BC MOE	Caribou camera research project	2018-2019
Masse Environmental	Mycology inventory	2017-2019
Selkirk College	Whitebark Pine remote sensing	2019
Transborder Grizzly Bear Project	Grizzly & Huckleberry research	2017-2019

Living Lakes Canada	Water quality monitoring sites	2019
C.Kootenay Invasive Sp. Society	Longterm invasive weeds monitoring	2018-2019
Simon Fraser University (MS Thesis)	Caribou habitat restoration	2018
BC Conservation Data Center	Plants at risk inventory	2017

2.2.2 Anti-Discrimination

Canada and British Columbia maintain a robust set of laws and regulations that are covered under various federal and provincial human rights and labour legislation. The Nature Conservancy of Canada is a prominent national organization that is compliant with the laws of Canada and B.C.. NCC also has well-developed internal human resources policies governing the management and conduct of NCC's employees: The Nature Conservancy of Canada Management Policy: Human Resources Policy (2012) and the NCC Darkwoods Occupational Health & Safety Program (2012). These policies include statements related to discrimination and related reporting and management practices. These policies are reviewed and updated by senior management annually, and all employees are trained and required to comply with these policies.

There have been no changes of note to these policies during the monitoring period.

2.2.3 Worker Training

NCC's policies related to health and safety, including safe operating procedures, are fully documented in: NCC Darkwoods Occupational Health & Safety Program (2012). Employees and contractors maintain programs that meet or exceed the B.C. Forest Safety Council requirements and WorkSafe B.C. Regulations. Other specific technical tasks such as carbon monitoring plot installation and measurement are trained at the start of each field season using standard operating procedure documentation.

There have been no changes of note to these policies during the monitoring period.

2.2.4 Equal Work Opportunities

See Section 2.2.2. There have been no changes to these policies during the monitoring period.

2.2.5 Workers' Rights

Canada has very well developed labour legislation governing workers' rights. Responsibility for the establishment of labour legislation generally falls under provincial jurisdiction, along with federal authority in certain industries. This legislation governs minimum employment standards that by law define and guarantee workers' rights within the workplace. Each province, as well as the federal government, has legislation governing workplace health and safety, as all Canadian workers have the right to work in a healthy and safe environment. Human rights provisions, fair employment practices and equal pay and anti-discrimination laws are covered under federal and provincial human rights and labour legislation.

In British Columbia, the B.C. Ministry of Labour, Employment Standards Branch is responsible for the administration of the provincial Employment Standards Act. This Act covers the following topic areas: Minimum Wage, Minimum Daily Pay, Meal Breaks, Paydays and Payroll Records, Overtime, Averaging Agreements, Uniforms & Special Clothing, Deductions, Statutory Holidays, Employing Young People Under 15, Compensation for Length of Service, Annual Vacation, Vacation Pay, Leave From Work, Temporary Foreign Workers, Collective Agreements and Resolving Disputes.

The Nature Conservancy of Canada has well-developed internal human resources policies governing the management of NCC's employees. The overarching Human Resources Policy is reviewed annually by senior NCC management and updated when deemed necessary. All NCC employees are informed of the existence of this policy on NCC's intranet website, which addresses the standards, requirements and procedures governing an employee's employment by NCC. All new employees are required to sign a disclosure indicating agreement and adherence to these policies and all NCC employees must renew this agreement on an annual basis.

Subcontractors, defined as service providers who bill NCC for their services, must sign an Agreement for Services contract, the standard template of which includes a separate provision entitled Compliance with Laws and Policies, as follows: "The Contractor shall be knowledgeable of and comply with the provisions of all applicable legislation, By-Laws of local authorities and Regulations in the Province of British Columbia and shall specifically ensure that the Contractor and his/her/its employees are knowledgeable of and performs all obligations imposed by the *Occupational Health and Safety Act* or equivalent in the Province of British Columbia".

In summary, NCC is a national organization fully compliant with the laws and regulations of BC and Canada. NCC employment policies and worker guidelines are documented in the latest versions of: The Nature Conservancy of Canada Management Policy: Human Resources Policy and the NCC Darkwoods Occupational Health & Safety Program.

There have been no changes to these policies or complicane during this monitoring period.

2.2.6 Occupational Safety Assessment

Darkwoods project activities are typical operations for forestry operations and timberland in BC, and include typical forest operations (logging, road work, etc.) and various forest field work. There are no unusual or unique activities, and therefore risks are typical of operations undertaken commonly across the forest industry.

In Canada, all 3rd party contractors operate as "independent contractors", a specific legal term that requires the contractors to operate and direct their delivery of services and employees independently. Therefore, operational safety procedures are the purview of each contractor. The WorkSafe BC regulations and BC Forest Safety Council SAFE Company programs are the primary methods in BC to assure worker safety in the forest industry, and include training and procedural materials used across the industry. Various levels of performance are self-audited by these programs.

NCC's policies related to health and safety, including safe operating procedures, are fully documented in: NCC Darkwoods Occupational Health & Safety Program. As per this document and other NCC annual plans, employees undergo various health and safety awareness and training programs, as applicable to their job function.

NCC contracts mandate that all contractors have a valid and active WorkSafe BC number. The contractor is then responsible to WorkSafe BC (https://www.worksafebc.com/en/about-us/who-we-are/mission-vision-values) to ensure that legal regulations are followed. Logging contracts also specify contractors are certified as BC Forest Safety Council SAFE Company.

NCC undertakes pre-work reviews with contractors as described in the NCC Darkwoods Occupational Health & Safety Program, which includes a pre-work safety site orientation checklist. NCC also ensures that logging contractors and other large contractors have at least equivalent procedures. Researchers are made aware of safety issues through both a research permit and waiver. And finally, general public access permits are also advised of the risks on the permits document.

There have been no changes to these policies or complicane during this monitoring period.

2.2.7 Feedback and Grievance Redress Procedure

The Nature Conservancy of Canada is a highly accessible organization and maintains multiple venues for contacting it, including:

- Offices with knowledgeable staff across the region in Nelson, Invermere, Victoria, Vancouver, and Toronto, ON
- Listed phone numbers for each site (phone book and website)
- NCC's website including address, phone number and email contact info for each BC office
- NCC's website's general "contact us" function
- Caretaker at Tye who interacts with Tye leaseholders and visitors and communicates directly to NCC
- Signs posted on the Darkwoods Property with NCC's toll free phone number

When NCC receives an expressed concern from a community member or interested interested stakeholder, NCC deals with it in the most appropriate manner. If the concern is addressed to NCC in writing, NCC will typically respond in writing as soon as possible (examples available upon request). In the more common instances in which NCC receive a phone call or in-person visit, concerns or comments are addressed in person. Regardless of which form of contact has been used, NCC has been able to satisfactorily address outstanding concerns in this manner. If a concern were to be elevated, NCC would engage a third party mediator if warranted or if, in certain instances, NCC is contractually obligated to do so.

NCC has a formal process to receive communications from interested stakeholders at local or regional offices, which are documented locally as received[1]. Any significant or unresolved issues are forwarded to the regional office, where records are kept and follow-up plans developed as necessary.

The B.C. Regional Office with delegated authority from the NCC National Office is the final authority for responding to any issues related to Darkwoods.

In general, local community members and interested stakeholders will contact local NCC staff for information or concerns via phone, email or social media.

There have been no material grievances or unresolved issues of note during this monitoring period.

2.2.8 Stakeholder Access to Project Documentation

Project documentation and updates are made available electronically via the NCC Darkwoods webpage and social media as needed. Interested stakeholders are generally directed to the Verra Project Database website for final project documentation, which it is permanently stored and available to any interested stakeholder. The Verra website documentation includes VCS, CCB, and SD VISta documentation. The public comment period for the draft SD VISta project documentation are made available via the Verra website and comment process, while NCC provides links and updates for stakeholders via the NCC website and social media accounts. Project documentation is also made available to interested stakeholders by NCC staff by request (this is not common). Generally, NCC and Darkwoods function within the modern electronic world where web and social media contact are the preferred communication methods for local communities and stakeholders.

2.2.9 Information to Stakeholders on Assessment Process

See Section 2.2.8. Additionally, NCC directly contacts key stakeholders by phone or email with notice of potential contact by the project assessor or during a site visit as appropriate.

2.3 Project Management

2.3.1 Avoidance of Corruption

The Nature Conservancy of Canada is a prominent national not-for-profit organization that complies with the laws and regulations of Canada and British Columbia, including those pertaining to corruption, etc. NCC maintains well-developed internal human resources policies governing the management and conduct of NCC's employees: See The Nature Conservancy of Canada Management Policy: Human Resources Policy (2012), and supporting documentation.

There have been no changes to this policy or issues related to it during this monitoring period.

[1] As of about 2013 NCC has completed operationalizing its new Land Information System, a robust internal data tracking system, within which calls and emails related to the Darkwoods property are now documented and retained digitally.

2.3.2 Recognition of Property Rights

NCC owns the Darkwoods property on a fee-simple basis, and all title and rights documentation is available for review. There are no title or property right disputes on Darkwoods.

2.3.3 Free, Prior and Informed Consent

No people live on or depend on the Darkwoods project area for their livelihood, and therefore this section is not applicable.

2.3.4 Restitution and/or Compensation for Affected Resources

No people live on or depend on the Darkwoods project area for their livelihood, and therefore this section is not applicable.

2.3.5 Property Rights Removal/Relocation of Property Rights Holders

No people live on or depend on the Darkwoods project area for their livelihood. No removal or relocation of property rights has occurred, and therefore this section is not applicable.

2.3.6 Identification of Illegal Activities

NCC maintains controlled (gated) access to the Darkwoods property, an access permitting process, and maintains a local staff presence to regularly monitoring the property area. There has been no evidence of material illegal activities on the Darkwoods property during this monitoring period.

2.3.7 Ongoing Conflicts or Disputes

There are no ongoing conflicts related to rights, lands, territories or resources related to the Darkwoods project area.

2.3.8 National and Local Laws and Regulations

The Darkwoods project has maintained compliance with all relevant laws and regulations during this monitoring period.

2.4 Grouped Projects

The Darkwoods project is not a grouped project.

3 BENEFITS FOR PEOPLE AND PROSPERITY

3.1 Impacts on Stakeholders

The Darkwoods project area is the private, fee simple property with gated road access control. There are no communities, indigenous people or other public communities or ongoing public use for income, livelihood, or critical cultural values by any people living or dependent on resources within the project area that would be affected by project activities during this monitoring period.

However, as described in the CCB PDD, the project calculates a series of quantitative and qualitative measeures of the impacts of the baseline vs. project scenario on local communities and interested stakeholders generally.

Implementation of the baseline scenario would have:

- Generated economic benefits for select community members due to the increased labour and income related to higher harvesting activity levels;
- Generated negative impacts by foregoing the investment and diversified economic activities in the project scenario, negatively impacting climate services, negatively impacting other ecosystem services (including water), and negatively impacting regional amenity values that are important to community members and community development.

As noted in the CBB PDD, Section G1 and G2.4, the project zone is a highly diversified modern economy that has a low dependence on forestry. Additionally, the forest industry makes up a relatively small component of the project zone economies, small component of project zone employment, and is a declining source of employment over the past several decades. During this monitoring period, the higher harvest levels in the baseline scenario would likely have expanded local forestry economic opportunities, but a substantial portion of these benefits would have accrued to stakeholders outside the project zone as a portion of services are provided by mobile outside contractors and a substantial volume of the harvested wood would have been delivered to external facilities[2].

[2] In addition, the CBB PDD discusses the fluid and flexible nature of regional log markets and regional log flows, and posits that on average the market would adjust to the presence or absence of the baseline harvest volume without spiking or growing employment in a direct linear manner. In reality, the regional log markets are very interconnected, and are more reactive to national and international market conditions than by small fluctuations in local log supply. We expect in reality the higher harvest levels in the baseline scenario would result in the market deferring higher cost and/or lower quality wood sources in the regional fiber basket (particularly on long term crown tenures), or otherwise changing log flows to be more cost optimal with the Darkwoods volume in the mix rather than just expanding to the supply. In other words, the potential benefits to community stakeholders in direct employment and economic opportunity would likely not expand in direct relationship to higher volume, but rather the market would substitute Darkwoods volume for other less optimal volume while keep production levels relatively similar. That said, to be conservative (and due to lack of direct supporting evidence) for this impact analysis we have assumed the entire Darkwoods baseline harvest volume contributes additive employment and economic opportunity to the project zone.

Comparatively, the implementation of the project has:

- Contributed direct economic development through project activities, but negatively impacted the economic benefits of the higher timber harvesting in the baseline scenario (due to lower harvest levels in the project);
- Generated substantial climate services benefits;
- Generated substantial other ecosystem services benefits, including water provisioning and quality, valuable fisheries contributions, potentially pollinator support, and other ecosystem services;
- Enhanced the community amenity values for the local region, including visual qualities, recreational opportunities, scientific study, and supporting regional environmental values and issues.

The following impact tables summarize the community impact monitoring for the project during this monitoring period. Additional background on local communities and these measures can be found in the projects CCB PD documentation.

Impact #1	Reduced economic opportunity from avoided timber harvesting.
Type of Impact	Negative, direct, estimated impacts. The avoidance of the higher levels of timber harvesting in the baseline scenario reduces the employment opportunities in the logging industry. The impact is estimated using regional economic modeling of timber related employment applied against the projected reduction in timber harvesting volume from the baseline scenario.
Affected Stakeholder Group(s)	Local and regional logging and log processing industry/workers.
Resulting Change in Well-being	Reduction in timber harvest volume during this monitoring period has a estimated net negative impact on economic opportunities in the logging industry, as shown in Table 4[3].

[3] See the project CCB PD documentation Section G2.4 for additional details on the baseline scenario impacts on communities, and related data and assumptions.

Table 4. Direct economic benefits in this monitoring period, by scenario

Project Year	Project Harvest Volume (m^3/yr)	Employment Generated (PY) Coefficient = 0.669	Employment Income ($) Avg. $42,630/yr
2017	-	-	$ -
2018	-	-	$ -
Avg/year	-	-	$ -
Total	-	-	$ -

Project Year	Baseline Harvest Volume (m^3/yr)	Employment Generated (PY) Coefficient = 0.669	Employment Income ($) Avg. $42,630/yr
2017	329,928	220.7	$ 9,409,372
2018	210,594	140.9	$ 6,006,029
Avg/year	270,261	180.8	$ 7,707,700
Total	540,522	361.6	$ 15,415,401

Impact #2	Increased economic opportunity from project activities.
Type of Impact	Positive, direct impact, predicted on an ex-ante basis, measured on an ex-post basis.
Affected Stakeholder Group(s)	Implementation of the project provides direct economic benefits to the local communities and across a wide range of stakeholders related to: 1. Project expenditures and activities 2. Project timber harvesting, silviculture, and operational activities 3. Improved recreational opportunities 4. Research opportunities
Resulting Change in Well-being	1. Project expenditures and activities - Table 5 shows NCC direct staff hours and spending that contribute to economic opportunities in the project scenarion in this monitoring period on an equivalent value basis as the logging in Table 4. 2. Project timber harvesting – although the project harvesting normally creates an economic benefit from the project, as shown in in Table 4 there was no harvesting in the project during this monitoring period.

Table 5. Additional economic benefits in the project scenario (non-harvesting)

	Total $	$/year	Person Year (@$42,630/yr)	Total Person Years (2017-2018)
NCC Staff Hours	$885,838	$221,460	5.2	10.4
Darkwoods Spending[4]	$1,035,162	$258,791	6.1	12.2
Total	$1,921,000	$480,250	11.3	22.6

[4] These include totals from NCC accounting records and relate to expenses such as forest research, management planning, carbon project costs, road deactivation and other expenses generally unique to the project scenario.

Impact #3	Climate services provisioning benefits.
Type of Impact	Positive and direct impact. Actual GHG emission impacts are measured and monitored, while quantitative value is estimated from various carbon market value sources.
Affected Stakeholder Group(s)	All local community members along with national and global benefits.
Resulting Change in Well-being	The ecosystems on the Darkwoods property provide significant climate benefits in the form of carbon storage and ongoing sequestration in the forest biomass. For this monitoring period, the benefits are estimated in Table 6.

Table 6. Darkwoods climate impact value by scenario

Net Emission Reductions by Scenario 2017-2018:

	Baseline (tCO_2e/year)	Project (tCO_2e/year)	Net Change (tCO_2e/year)
2017	(325,881)	109,873	435,754
2018	(329,490)	73,669	403,159
Period Total	(655,371)	183,542	838,913
Period Average per Yr	(327,686)	91,771	419,457

Carbon Pricing Scenarios ($CDN):

	BC Legislation	Voluntary	CA ARB
Avg. $CDN/tCO2e	$30.00	$11.34	$14.63

Climate Services Value Estimate Scenarios (Avg. $CDN/Year)

	Carbon Price ($CDN/$tCO_2e$)	Baseline Net Emissions ($/year)	Project Net Emission Reductions ($/year)	Net Climate Benefit ($/year)
BC Carbon Tax Scenario	$30.00	$(9,830,565)	$2,753,130	$12,583,695
Voluntary Market Scenario	$11.34	$(3,715,954)	$1,040,683	$4,756,637
CA ARB Scenario	$14.63	$(4,794,244)	$1,342,667	$6,136,911
Avg. Value Impact ($/yr):		$(6,113,587)	$1,712,160	$7,825,747

Impact #4	Ecosystem services provisioning benefits.
Type of Impact	Positive and generally direct impacts. Impacts are estimated. Implementation of the project has positive impacts on: 1. Water Quality/Quantity 2. Key Species Habitat 3. Pollinator Services 4. Recreational Hunting/Fishing 5. Outdoor Recreation 6. Wetland Services/Protection These impacts can be qualitatively estimated based on regional data and studies, as shown in **Error! Reference source not found.**.
Affected Stakeholder Group(s)	All local community members along with national and global benefits.
Resulting Change in Well-being	This suite of ecosystem services provisioninig are evaluated on a qualitative basis for the project. During this monitoring period the assessed results are unchanged, as shown in Table 7.

Table 7 - Qualitative Assessment of Non-Climate Ecosystem Services Provision Impacts.

Ecosystem Service	Baseline Impacts	Project Impacts	Magnitude of Value
Water Quality/Quantity	-	++	Significant[5]
Key Species Habitat	--	++	Significant[6]
Pollinator Services	-	+	Low-Moderate[7]
Recreational Hunting/Fishing	+[8]	++[9]	Low to Significant
Outdoor Recreation	-	+	Moderate[10]
Wetland Protection	-	+	Low[11]

[5] In particular, the Darkwoods property contributes significantly to the valuable Kootenay Lake fishery, worth $5-10 million per year to the local community (page 8: http://www.env.gov.bc.ca/kootenay/fsh/main/pdf/KLAP%20Kootenay%20Lake%20Action%20PLan%20final%209_May_2016.pdf).

[6] In particular, the endangered mountain caribou, grizzly bears, wolverines and other key species with extensive regional planning, community involvement, and public costs. The Darkwoods project significantly contributes habitat, migration corridors, and other values to these and other species. This contributes to the success of these regional efforts and directly reduces short and long term community costs related to research, monitoring, habitat protection, and public land allocation/acquisitions.

[7] As discussed in the CBB PDD, fully functional ecosystems in the project scenario likely support native pollinators, which may have some positive impact on local Creston Valley agriculture, either through direct pollination services or more likely contributes to species diversification, stabilizing closer populations, and providing replacement populations as necessary.

[8] Some key game species (i.e. deer) may benefit from improved browse habitat created by logging in the baseline scenario, although extensive logging may reduce other habitat attributes for such species. It is unclear whether access for hunting in the baseline scenario would have been allowed, as regional property practices are mixed. To be conservative, a positive impact rating is given.

[9] The project would also benefit key game species such as deer and elk (browse in harvested areas, winter cover in retained forest, predation cover, reduced road access, etc.). The benefit to recreational fishery on Kootenay Lake may be significant, as undisturbed Darkwoods stream systems contribute directly to water quality, quantity, temperature, invertebrate populations, and fish hatchling poputations into Kootenay Lake and other watersheds. NCC has allowed public hunting on parts of the property (whereas the previous owner excluded hunting) which is an additional benefit to local community.

[10] Darkwoods provides direct benefits to 200-350 backcountry recreational users of the property each year. It is doubtful the fully operational and heavily logged property in the baseline scenario would provide a similar recreational draw.

[11] Significant wetlands are generally protected by minimum legal operating practices, and hence would have minor impacts from either scenario; however the drainages into and out of wetlands and small wetlands may be impacted by logging in the baseline scenario, leading to a negative overall impact rating.

Impact #5	Community amenity value benefits.
Type of Impact	Impact is positive, indirectly created, and estimated.
Affected Stakeholder Group(s)	Local communities (all).
Resulting Change in Well-being	The local communities place a high value on the environment and related recreational and quality of life values. The Darkwoods project contributes to these values and improves the quality of life, livability, and desirability of living in the local area (i.e. a qualitative positive impact). There have been no published changes to community values during this monitoring period.

3.2 Stakeholder Impact Monitoring

Community Impact Monitoring Plan

The primary objective of the Darkwoods community monitoring is to ensure that the project continues to provide the key employment, climate services, ecosystem services provisioning, and community amenity values as described in Section CM.

Community Impact Monitoring Plan:

Tracked annually and reported for each verification period.

Monitoring Activity 1 - Economic/employment Impact:

Objective: Track and report on NCC economic investment and spending related to the Darkwoods project implementation. Demonstrate that NCC continues to provide diversified economic benefit to the community and region through project related activities and monitoring.

Tasks:

1. Generate and summarize NCC accounting records for expenses related to Darkwoods.
2. Analyze comparative employment value estimates between the baseline scenario and the project expenses during the current verification period.

Monitoring Activity 2 – Climate Services Impact:

Objective: Track and report on the achieved net emission reductions generated by the Darkwoods project in the verification period.

Tasks:

1. Report on the most current ex-ante projections for net emissions from the relevant baseline and project years, and updated relevant average carbon value measures (i.e. B.C. carbon tax and/or offset pricing; CA ARB or Quebec allowance pricing, voluntary market sources, or other justifiable sources).
2. Report on the VCUs projected and issued, along with approximate sales proceeds and prices (to be provided on a confidential basis to verifiers only).
3. Calculate an average estimated climate services value for the verification period.

Monitoring Activity 3 – Other Ecosystem Services Provisioning Impact

Objective: Qualitatively (or quantitatively, if relevant data is available) re-assess the provision of other ecosystem services provisioning by the project in the verification period in comparison to an analysis of the potential baseline scenario impacts.

Tasks:

1. Report on any material project area landuse changes that impact key ecosystem services provisioning
2. Update qualitative value assessments for other ecosystem services in the baseline and project scenario years relevant to the verification period.

Monitoring Activity 4 – Community Amenity Value Impacts

Objective: Qualitatively (or quantitatively, if available) re-assess the provision of positive community amenity value from the project in the verification period in comparison to an analysis of the potential baseline scenario impacts.

Tasks:

1. Report on any material project area landuse changes that impact positive community amenity value.
2. Report on any updated community quality of life or other survey or community information that reports on the value of surrounding area environmental values (for example, references to the importance of water & air quality, recreation, scenery/aesthetics, etc.).

Key Outcome: demonstrate that, on average, the project continues to deliver net positive community benefits in comparison to the baseline scenario by updating the key quantitative and qualitative value assessments made in Section CM1, within each verification period.

All of these monitoring tasks have been completed for this monitoring period, as reported in the impact tables above.

3.3 Net Positive Stakeholder Well-being Impacts

The project has had a net positive impact for local communities and intersted stakeholders during this monitoring period. The project has a net negative impact on direct economic opportunities, with project spending less than the comparable estimated economic opportunity for the community with logging. However, the project generates substantially more value from climate and ecosystem services provisioning and community ammenity value on a quantitative and qualitative basis. The net impacts for this monitoring period are summarized in Table 8.

Table 8. Summarized Net Community Benefits

Project Zone Net Impacts (total monitoring period, 2017-2018)

	Baseline Scenario				Project Scenario			
	Activity (Person Years)	Economic Impact ($CDN)	# Positive Impacts	# Negative Impacts	Activity (Person Years)	Economic Impact ($CDN)	# Positive Impacts	# Negative Impacts
Employment	361.6				22.5			
Economic Benefit		$ 15.42				$ 1.92		
Climate Services Value		$ (17.13)				$ 4.80		
Sub-Total Economic Value:		$ (1.72)				$ 6.72		
Ecosystem Services (net count)			1	-6			9	0
Amenity Servies			0	-1			1	0
Net Services Impact Total:				-6 (-)				10 (+)

In total, the project provides net positive benefits to People and Prosperity through economic opportunities, climate impact value, and various ecosystem services provisioning value.

4 BENEFITS FOR THE PLANET

4.1 Impacts on Natural Capital and Ecosystem Services

NCC has identified key biodiversity and natural capital targets for Darkwoods that are monitored and reported in annual Effectiveness Monitoring (Table 9 and Table 10)[12]. Overall property management and biodiversity targets are described in the Darkwoods Property Management Plan, which follows adaptive management principals as it is updated every five years. The primary and current key indicators of effectiveness monitoring include:

Impact #1	Dry interior cedar-hemlock forest
Type of Impact	Positive, actual, direct impacts. Old seral stage dry ecosystem forests (ICHdw) are important to key habitat and ecosystem function on Darkwoods, and hence managing to maintain or increase the proportion of the property area in these old forest conditions is important to biodiversity values.
Affected Natural Capital and/or Ecosystem Service(s)	Proportion of ICHdw that is > 140 years old.
Resulting Change in Condition	Net positive – see Table 10.

Impact #2	Grizzly Bear
Type of Impact	Positive, actual, direct impacts. Grizzly Bears are important capstone species regionally.
Affected Natural Capital and/or Ecosystem Service(s)	1. Proportion of core in Structural Stage 3 2. Proportion of core with no access/non-motorized access
Resulting Change in Condition	Net positive – see Table 10.

[12] Note also the Benefits for the Planet that are quantified as part of the Benefits to People in section 3.

Impact #3	Mountain caribou
Type of Impact	Positive, actual, direct impacts.
Affected Natural Capital and/or Ecosystem Service(s)	Proportion of capable habitat lost due to anthropological means.
Resulting Change in Condition	Net positive – see Table 10.

Impact #4	Hydro-riparian ecosystems
Type of Impact	Positive, actual, direct impacts.
Affected Natural Capital and/or Ecosystem Service(s)	1. Barriers to fish flow/passage on the property 2. Proportion of watersheds with Equivalent Clearcut Areas of <20% 3. Number of bridges on the property at high risk of failure 4. Kilometers of roads rates at medium or high risk of failure
Resulting Change in Condition	Net positive – see Table 10.

Impact #5	Old growth cedar-hemlock forest (ICH types)
Type of Impact	Positive, actual, direct impacts. Old seral stage ecosystem cedar-hemlock forests are important to key habitat and ecosystem function on Darkwoods, and hence managing to maintain or increase the proportion of the property area in these old forest conditions is important to biodiversity values.
Affected Natural Capital and/or Ecosystem Service(s)	Proportion of ICH that is > 140 years old.
Resulting Change in Condition	Net positive – see Table 10.

Impact #6	Rare ecosystem and features
Type of Impact	Positive, actual, direct impact.
Affected Natural Capital and/or Ecosystem Service(s)	Proportion of rare ecosystems and features protected.
Resulting Change in Condition	Net positive – see Table 10.

Impact #7	Invasive species
Type of Impact	Risk, predicted, direct. Invasive species present a risk to natural ecosystems and hence the project maintains an active invasive species monitoring and removal program.
Affected Natural Capital and/or Ecosystem Service(s)	Number of invasive species occurrences and removals on the property. .
Resulting Change in Condition	Net positive - see Figure 3.

No known invasive species have been introduced to Darkwoods as a result of project activities. Invasive species occurrence data are available from the Invasive Alien Plant Program (IAPP) (https://catalogue.data.gov.bc.ca/dataset/invasive-alien-plant-site), run by the BC Government's Ministry of Forests, Lands and Natural Resource Operations (FLNRO). A total of 8 species have been recorded during the previous monitoring periods on the project area, and the number of species has not been increasing during the project. NCC maintains an active invasive species removal program, including . monitoring for the the presence of invasive species on a periodic basis across the property and removal of known invasive species on a periodic or annual basis by chemical, biological, or mechanical means. Figure 3 shows the occurrence of invasive species removal activities during a previous monitoring period, and is represented of the activities made in this monitoring period. .

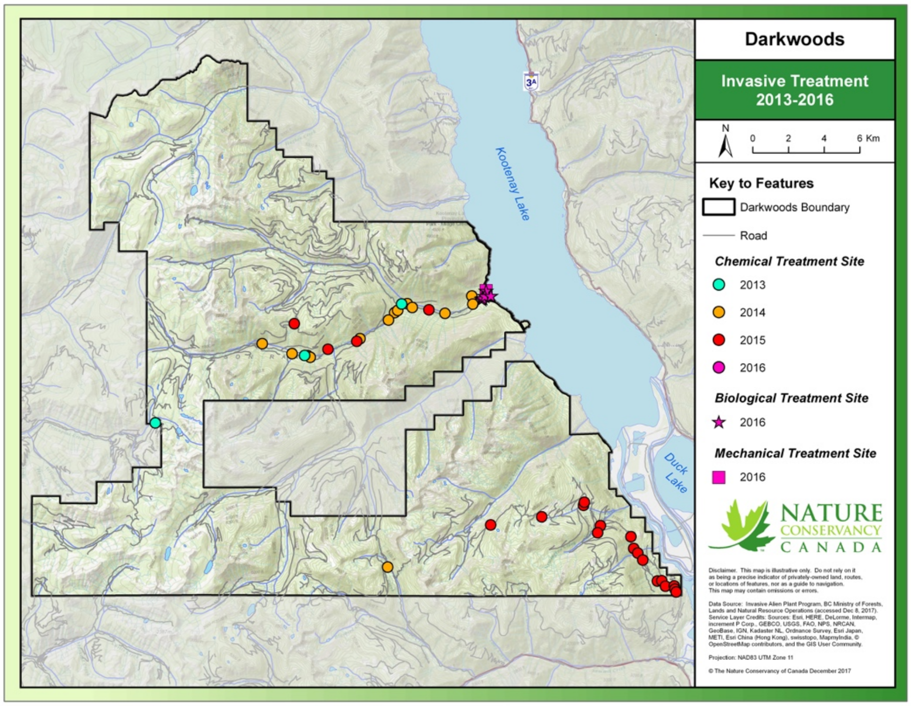

Figure 3. Darkwoods Invasive Species Removal Program 2013-16

Note also the impacts on Natural Capital and Ecosystem Services discussed in Section 3 above, in particular the estimation of a quantitative and/or qualitative assessment of climate and ecosystem services provisions outlined in Impact #3 and Impact #4 (Section 3 Impact Tables).

4.2 Natural Capital and Ecosystem Services Impact Monitoring

Biodiversity monitoring activities and processes on the Darkwoods property are designed to provide sufficient information to reduce threats and meet biodiversity goals, and to be adaptive to new information, objectives, and targets over time. Detailed discussion of the biodiversity threats, targets, and related management objectives and activities are described in the Darkwoods Property Management Plan (PMP) (NCC 2016), and as shown in .

The current targets and activities for biodiversity monitoring are termed “Effectiveness Monitoring”. The current project Effectiveness Monitoring is summarized in **Error! Reference source not found.** and the most recent reported results shown in Table 10.

As per the Darkwoods CCBA PDD, the management of biodiversity is outlined in the Darkwoods Conservation Area Property Management Plan 2017-2021 (NCC, 2016) for this monitoring period, as summarized in Table 17. The NCC Property Management Plan (2016) document provides detailed descriptions of the biodiversity effectiveness program, including background, references and assumptions. Consistent with this plan, biodiversity effectiveness monitoring is undertaken as before, now on a 5 year basis to better reveal changes and trends in long term habitat and feature development. .

Table 9. 2016 Darkwoods Property Management Plan biodiversity targets (summarized, see Table 7, Darkwoods PMP 2016).

Biodiversity Target	Ecological Justification	Size Impact	Property Condition	Landscape Context	Viability Rank
Carbon Sequestration	Mitigating the impacts of climate change	Very Good	N/A	Good	Very Good
Fire Maintained Ecosystems	ICHxw and ICHdw ecosystem are rare, degraded, support many species at risk	Very Good	Good	Fair	Good
Hydroriparian Ecosystems / Wetlands	Rare, diverse ecosystems. Support key fish populations	Very Good	Good	Very Good	Good
Mid-elevation Forests	Old Growth Interior Cedar-Hemlock.	Very Good	Good	Good	Good
Whitebark Pine Ecosystems	Keystone species in high elevation forests. Endangered.	Good	Poor	Good	Fair
Woodland Caribou Habitat	Critical habitat, highly endangered trans=border herd.	Poor	Poor	Fair	Poor
Grizzly Bear	Key habitat for important sub-population. Umbrella species	Good	Good	Good	Good

As of the 2016 Property Management Plan, NCC has shifted to reporting the biodiversity effectiveness monitoring data on a 5 year increment to better show long term changes. The previously reported annual data from 2013-2016 are represented of the previous data and trends, and is expected to be representative (at minimum) of the results and trends that will be found at the next monitoring period. As shown in , the project area has maintained or improved the key biodiversity effectiveness monitoring indicators representative of this monitoring period (Table 10).

Table 10. Summarized Annual Biodiversity Effectiveness Monitoring Results (2013-2016 period – next monitoring data reported 2021).

Target	2013	2014	2015	2016	Methods	Source
Dry interior cedar-hemlock forest						
Proportion of old forest in ICHdw	0.09	0.09	0.09	0.09	Total forested ICHdw1 = 4328.5 ha, Total ICHdw1 > 140 yrs = 397 ha OR 9% **Forested = Spc1 not blank, remove age < 3 (NSR). No harvest within this defined area in this period.	Used Ecora VRI, Deb MacKillop BGC, GIS harvest data.
Grizzly bear						
Proportion of core in Structural Stage 3 (SS3)	0.24	0.24	0.24	0.24	strstg3 = 6244/26211 = 24%or Core is SS3	Used Ecora VRI Struct Stg
Proportion of core no access/non-motorized	0.87	0.87	0.87	0.87	Of the 26,211 ha of core mapped habitat, 22830 ha is zoned No Access or 87% of Core is No Access	Darkwoods boundary, Primary Zone expanded
Mountain caribou[13]						
Proportion of capable habitat lost	0.00	0.00	0.00	0.00	No logging in caribou habitat areas in the monitoring period	GIS annual harvest data

[13] Also see: Footnote **Error! Bookmark not defined.**.

to disturbance						
Hydro-riparian ecosystems						
Barriers to fish flow/passag e	0.00	0.00	0.00	0.00	No new activity in period	Staff reporting of activities
Proportion of watersheds with Equivalent Clearcut Areas of <20%	1.00	1.00	1.00	1.00	See annual EM report worksheets	Used carbon VRI resultant
Number of bridges on the property at high risk of failure	0.00	0.00	0.00	0.00	6 Bridges removed since 2009, 1 installed	Staff reporting of activities
Kilometres of roads rated at medium- or high risk of failure	0.00	0.00	0.00	0.00	Up to Fall 2015 - 218 km deactivated out of 603 total km of road	Staff reporting of activities
Old-growth cedar-hemlock forest						
Proportion old forest in ICH	0.28	0.28	0.28	0.28	Total forested ICHmw4 = 5638.3 ha, Total ICHmw4 > 140 yrs = 1601 ha OR 28% **Forested = Spc1 not blank, remove age < 3 (NSR). No harvest or fires within this defined area in period.	Used Ecora VRI, Deb MacKillop BGC, GIS harvest data.
Rare ecosystems and features						
Proportion of sites protected	1.00	1.00	1.00	1.00	No new activity in period	Staff reporting of activities

In addition, as part of the CCB PD, NCC monitors Invasive Species, as displayed in Figure 3.

4.3 Net Positive Natural Capital and Ecosystem Services Impacts

In addition to the net positive impacts from ecosystem services noted in Section 3, the project will have net positive impacts on biodiversity as shown in section 4.2 as well as significant GHG emission reductions as shown below.

The results of biodiversity effectiveness monitoring in the project periods to date have demonstrated the project has positive direct impacts on all of the key indicators described in Section 4, and shown in Table 10.

Additionally, the project has validated and verified GHG emission reductions and forest biomass protection and growth as demonstrated in the VCS PD and Monitoring Reports verified to date, and summarized below. These significant climate related benefits will continue throughout the project lifepan.

Summary of GHG Emission Reductions:

The Darkwoods project area sequesters significant carbon stocks in the forest biomass on the property. The avoidance of baseline scenario logging results in the annual retention of carbon stocks by avoiding the removal of tree biomass and subsequent decay of the resulting dead biomass and wood product waste. The site also continues to sequester additional carbon in the growth of the retained and regenerating forests across the site. These carbon stocks and their fluxes overtime have undergone detailed measurement and modeling by carbon pool, using the Verified Carbon Standard (VCS) Methodology VM0012, as documented in the Validated Darkwoods Forest Carbon Project Design Document (PDD) and as verified in subsequent VCS Monitoring Reports.

The following is a summary of the key data and monitoring activiteis related to GHG emission reductions during the monitoring period:

The Darkwoods project monitoring plan contains three primary activities:

1. Annual inventory change monitoring

NCC has undertaken a non-systematic spatial monitoring on the entire property annually over the monitoring period, and the GIS inventory is updated annually. This was undertaken via a combination of aerial helicopter reconnaissance, field observations, field surveys of identified project activities, and field surveys of any identified disturbance events (i.e. fires, etc.). In addition, NCC has updated inventory databases with information from biodiversity and other non-carbon monitoring programs installed since 2008.

a. Monitoring of natural disturbance events:

NCC has collected monitoring data on the spatial extent of annual forest fires during the monitoring period. There were no material fires on Darkwoods in 2017, and two fires in a significant fire season in 2018. These two fires initiated outside the project area and burned into the property, one from the central Next Creek property which burned ~9 ha on Darkwoods, and a fire from adjacent lands to the south of the property which burned 2,438 ha on Darkwoods, for a total of 2,447 ha burned.

The burned area was mapped by helicopter GPS by the BC MOF, as shown in Figure 4. Additionally, the BC MOF has begun a new fire tracking system which also evaluates and maps burn severity. Therefore, for the 2018 fires, the project has new details to more accurately account for both carbon emissions and resulting forest stand conditions after the fire. Interestingly, this severity tracking reveals that although the fires covered a relatively large area, there were only small areas of severe burn damage, while as significant area was classified as unburned, as shown in Table 11. The project has utilized this burn severity data to improve the accuracy of emissions impact. The history of forest fires on the property during the project lifetime is shown in Table 12

Table 11. 2018 Fire Summary Data

Burn Severity Class[14]	GIS Hectares	% of Fire Area	Modeling Assumption
Low	379.3	15.5%	No impact on biomass stocks[15]
Medium	846.5	34.6%	Equivalent to Clear-cut[16]
High	200.1	8.2%	Equivalent to Clear-cut
Unburned	1021.4	41.7%	No impact to biomass stocks
Unknown	0	0%	Not Applicable
Total	2,447.3	100%	

[14] Burn Severity Class descriptions can be found in the Post-wildfire Natural Hazards Risk Analysis in British Columbia: https://www.for.gov.bc.ca/hfd/pubs/docs/lmh/lmh69.pdf

[15] Generally this assumption is that these stand areas will remain similarly stocked and did not, on average, lose significant amounts of tree biomass volume; and any loss will likely be recovered rapidly due to nutrient flushing, reduced competition, etc.

[16] For simplicity, the medium and high damage fire areas will be treated the same as clear-cut stands, with all HWP volume being emitted immediately. Severe fires are infrequent enough on Darkwoods that further refinement of post burn stand conditions and modeling is more detail than required. This assumption is conservative, as even the high burn severity retain substantial amounts of biomass in burned snags and stumps which would emit carbon on a much longer timeframe than an assumed clearcut, meaning this assumption does not increase claimed emission reductions.

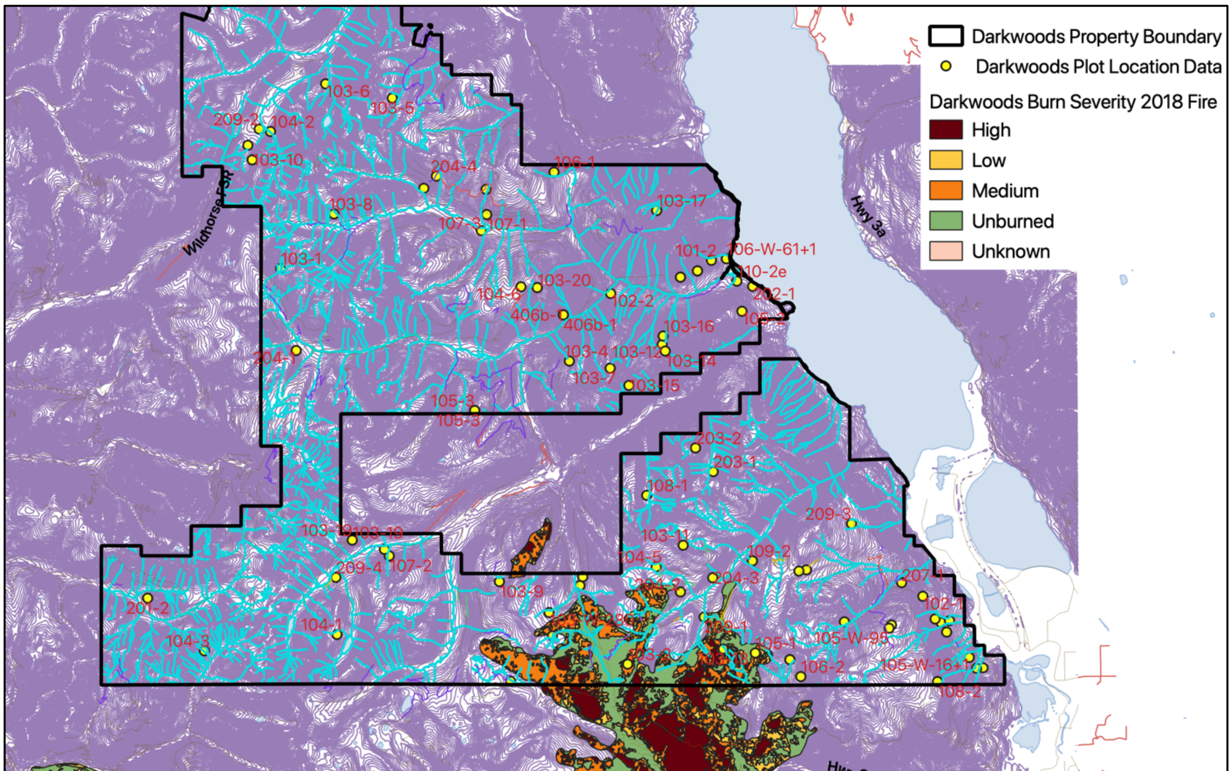

Figure 4. 2018 Fire - burn severity rating map

Table 12. Darkwoods monitored fire events since project inception.

Project Year	# Fire Occurrences	Monitored Fire Occurrence (ha)
2008-2013	0	0 ha
2014	4	80.8 ha
2015	1	102 ha
2016	0	0 ha
2017	0	0 ha
2018	2	2,447 ha

2. Planned project activities (i.e. harvests, road construction, reforestation, etc.)

NCC did not undertake any timber harvesting or other activities materially affecting carbon stocks in this monitoring period. Generally NCC looks for opportunities to deploy limited timber harvesting to ways that are both economical and in support of property management and conservation objectives. In evaluating the timber harvesting opportunities and management plans in 2017 and 2018, NCC decided not to conduct harvesting. NCC expects to harvest timber in limited amounts on an annual or periodic basis as per the ex-ante project plan in future monitoring periods.

Table 13. Darkwoods Project Harvesting Activities 2017-2018

Project Year	Year	Ex-Post Projected Volume (m^3)	Ex-Post Delivered Volume (m^3)	Ex-Post Actual Area (ha)
10	2017	0	0	0
11	2018	0	0	0
	Total	0	0	0

3. Unplanned man-made disturbances (i.e. non-de minimis illegal or unplanned harvests, etc.):

No unplanned or illegal harvest or other activities were observed on the property during the monitoring period, thus no inventory updates or adjustments have been necessary. NCC has completed an access management plan, installed access controls on the property (i.e. gates), and implemented a mandatory permit system to monitor access on the property.

4. Other Monitoring Requirements of the Project

NCC also documented other monitoring requirements of the PDD, including:

a. Activity shifting leakage (annually)

NCC has not undertaken any material level of timber harvesting on any owned or managed property outside Darkwoods for the period 2017 and 2018, and therefore there is no risk of starting activity shifting risk (see Table 14). Records of commercial activities are kept by NCC for each property in an in-house online database and accounting records.

Table 14. NCC Logging Activity on Other Properties (activity shifting risk) 2013-2016.

Property	Project Year	Logging Volume (m^3)	Activity shifting evidence/comment
All NCC properties outside Darkwoods	2017	0.0	n/a – no harvest
All NCC properties outside Darkwoods	2018	0.0	n/a – no harvest

b. Annual market leakage calculations (annually)

Annual calculations of market leakage have been made, resulting in a leakage deduction rate of 20% for this monitoring period.

c. Field plot monitoring

i. Sampling: Carbon Plots and Re-measurement

The project has installed a total of 71 active permanent sample plots for carbon. Permanent plots are re-measured at intervals of not longer than 5 years (beginning at their date of first measurement, and interpreted as 5 field seasons instead of calendar years). Sampling in the project landbase has been stratified such that each forest analysis unit has plot representation in proportion to its cumulative productive area in the project landbase (See Table 5).

In 2017 the first set of plots were re-measured on the 5 year increment. These re-measured plots were verified as part of the 2013-2016 verification audit. Subsequently the remaining 12 plots have been remeasured in 2019. In 2019 a plot was burned in the 2018 forest fire, and was re-established in 2019 at the same location. Other than this replacement plot, no new plots have been added during this monitoring period. The inventory error term sits at 10.9% and the project error term sits at 3.0% suggesting that the project has the appropriate number of plots to achieve error targets (net change in carbon stocks within 10 percent of the true value of the mean at the 90-percent confidence), for the time being.

All plot locations, types, tree data and CWD data are stored in an excel spreadsheet provided to verifiers for review.

Plot Data Analysis Summary

The project uses this carbon plot data to monitor both the development of carbon stocks over time and to monitor the accuracy of modeled carbon stocks by analysis unit; which then is adjusted using the uncertainty factor statistical analysis and deductions. The project calculates actual carbon content on each plot on a tC/hectare basis using the tree and CWD data collected on each plot. Plot tree data is input directly as stand inventory information into the FVS model on a plot by plot basis, which then outputs above- and belowground live, and standing dead carbon on a tC/hectare for each carbon pool. The CWD transact data is calculated separately in the project Access database following equations to convert transect measured data into tC/hectare.

This total plot ecosystem carbon data is then used to compare actual carbon stocks on an analysis units basis to the modeled carbon stocks on each AU. These data are compared in the uncertainty factor calculation, which results in a deduction based on the accuracy of the inventory and modeling data.

Quality Assurance/Quality Control Measures (QA/QC)

The QA/QC standards described in the PD were adhered to during this monitoring period::

QA/QC for Field Measurements conducted during the 2017-2018 monitoring period

After the plot remeasurements in 2017 and 2019 NCC undertook blind check cruises for ~10% of the plots. Check plots were undertaken by experienced field staff or Darkwoods staff on a blind basis. program, which resulted in 8 plots being re-measured. DBH, Height, and Age measurements met the accuracy threshold of ±10% at 90% CI, as required by the PDD.

QA/QC for Data Entry for 2017-2018 monitoring period

NCC standard procedures and ongoing QA/QC programs for data were followed to ensure proper entry of data from paper to electronic format (see SOP document ‘Darkwoods Data Standards_28Aug17.doc’). The primary quality mechanism in the project is electronic data recorders with field based entry followed by electronic file transmission for further data analysis. No errors were noted during analysis for this verification period.

QA/QC for Data Archiving for 2017-2018 monitoring period

NCC has stored the electronic field data files and summary excel files on NCC servers and in Dropbox project folders, and age core samples are retained at the Darkwoods field office (see SOP document ‘Darkwoods Data Standards_28Aug17.doc’).

Quantification of GHG Emission Reductions and Removals

For the 2017-2018 monitoring period, baseline emissions were calculated using the same process outlined in the PD and previously used in prior verifications, with a few exceptions noted below.

The GIS database has been updated for project activities and stand ages for the years in this monitoring report. There have been no changes to this GIS inventory/landbase stratification an analysis process in this monitoring period.

For this monitoring period, the project has transitioned from the FORECAST and FPS-ATLAS models to FVS_FFE and the FPS harvest scheduling model. The FVS_FFE model is widely used, understood and accepted for use across both forestry and forest carbon project applications. It is likely the most common forest model supporting forest carbon projects across North America. The combination of FVS_FFE and FPS meet all six criteria for model selection in the VM0012 methodology. Although similar to FORECAST as a forest growth and yield models, the FVS_FFE model further automates forest carbon modeling with the integrated modeling of all carbon pools and harvested wood products using methods that follow the VCS and VM0012 requirements.

Baseline Emissions

The baseline scenario harvesting has been modeled as described in the VCS/CCB PD and Monitoring Reports. The carbon stock changes related to this baseline harvesting have been modeled for the monitoring period using the FVS_FFE model and related database tools. The volume harvested and the resutling annual net emissions/reductions for the baseline scenario are summarized in Table 15. Additional details related to GHG emission reduction calculations can be found in the Darkwoods VCS/CCB Monitoring Report 2017-2018.

Table 15. Annual emissions/reductions for the baseline scenario calculated for 2017-2018.

Year	Ann. Baseline (Emissions) Reductions (tCO2e)	Volume Harvested (m^3)	Baseline Storage HWP (tC)	Baseline Storage Waste Products (tC)	Baseline (Emissions) Production & Waste (tC)
2017	(325,881	329,928	4,208	1,356	(4,224)
2018	(329,490)	310,594	3,961	1,276	(3,977)

Project Emissions

The project activities materially affecting forest biomass and carbon stocks have been modeled as described in the VCS/CCB PD and Monitoring Reports. The carbon stock changes related to the project activites have been modeled for the monitoring period using the FVS_FFE model and related database tools. The volume harvested and the resutling annual net emissions/reductions for the baseline scenario are summarized in Table 15. Additional details related to GHG emission reduction calculations can be found in the Darkwoods VCS/CCB Monitoring Report 2017-2018.

For the 2017 – 2018 monitoring period, no timber harvesting occurred on the property.

The fire areas were overlayed on the forest inventory and new polygons created for the medium and high burn severity areas, which are then identified as burned, but treated as clear-cuts for the purpose of carbon inventory and modeling. These new polygons then follow the carbon analysis unit transition process to regenerating stands as normal timber harvesting. The clear-cut equivalent method is the same as used previously on other fire areas on Darkwoods, however in this case we are now stratifying the fire area based on severity. The no impact and low severity areas are marked as fire, but retained as the original stand characteristics. The net harvesting volume and project emission reductions are show in Table 16.

Table 16. Annual emissions/reductions for the project scenario calculated for 2017-2018.

Year	Ann. Project (Emissions) Reductions (tCO2e)	Volume Harvested (m^3)	Project Storage HWP (tC)	Project Storage Waste Products (tC)	Project (Emissions) Production & Waste (tC)
2017	109,873	0	0	0	0
2018	73,669	0	0	0	0

Net GHG Emission Reductions and Removals:

The net changes to emission reductions between the baseline and project scenario *are* shown in Table 17.The application of key deductions and buffer contributions to calculate claimed VCU's is shown in and Table 18.

Table 17. Summary of emission removals and leakage emissions for the baseline and project scenarios for the 2017-18 verification period.

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
2017	(325,881)	109,873	(87,150)	348,604
2018	(329,490)	73,669	(80,631)	322,522
Total	(655,371)	183,542	(167,781)	671,126

Calculation of Claimed Verified Carbon Units (VCU's):

The application of key deductions and buffer contributions to calculate claimed VCU's is shown in Table 18, with the following deductions made:

1. Leakage = 20%.
2. Uncertainty Factor = 1.5%.
3. Non-Permanence Risk Buffer Contribution Rate = 10%

Table 18. Total issuable VCUs and terms used in their calculation for the 2017-2018 verification period.

Year	Net GHG emission reductions or removals (tCO_2e)	Uncertainty Risk Discount (tCO2e)	Annual Buffer Set-Aside (tCO2e)	Issueable VCU (tCO2e)
2017	348,604	(5,199)	(34,860)	308,545
2018	322,522	(4,838)	(32,252)	285,432
Total	671,126	(10,037)	(67,112)	593,977

Overall, the project achieves net positive impacts for the Planet through biodiversity and climate related elements, in addition to the ecosystem services provisioning impacts identified in Section 3.